

Network-Augmented Latent Factor Portfolios for Cryptocurrencies

Using Time-Varying Community Structure as Instruments for IPCA Pricing

Artemis Quant Competition Submission — 2026 cycle

Track 1: Crypto Factor Rebalancing Strategy

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Abstract

We propose NALFP, a synthesis of three previously-disconnected strands of cryptocurrency research: time-varying community detection on correlation graphs, latent-factor asset pricing, and adaptive portfolio construction. The novel contribution is to feed two time-varying network signals — the within-cluster momentum rank and the cross-cluster relative strength — into the characteristic matrix of an Instrumented PCA (Kelly et al., 2019) model as additional instruments alongside the standard market, momentum, volatility, size, fundamental yield and supply absorption factors. The IPCA expected returns are then blended with a pure network signal using an out-of-sample IC-proportional weight, and the resulting cross-sectional rank drives a weekly-rebalanced long/short quintile portfolio with explicit cluster-diversification, single-asset, turnover and volatility-target constraints. On 16 out-of-sample weeks the constrained strategy delivers a Sharpe of 1.66 (95% block-bootstrap CI $[-0.71, 7.11]$) at 5.6% realized annual volatility and 8.7% weekly turnover; an intentionally-uncapped ablation reaches a Sharpe of 3.15 (CI $[1.34, 7.27]$) on the same window. We document — and quantify — where the strategy breaks down: zero IPCA latent factors clear the $|t| \geq 1.65$ filter on 24 training weeks, the cluster cap actively reduces Sharpe in this slice of history, and three of nine instrument signs in Γ are inconsistent with prior crypto factor evidence. The submission is offered as an exercise in methodological honesty rather than as a claim of identified alpha.

Competition track. 1 — Crypto Factor Rebalancing Strategy.

Code and data. Reproducible from `08_nalfp/` on the project repository; the full pipeline runs in under three minutes.

1 Introduction

The Artemis Analytics competition asks for a systematic, data-driven rebalancing strategy with two requirements: the factor must be economically motivated, and the team must be able to articulate the limits of its own approach (Artemis, 2026). We take both seriously. Our methodological starting point is that three published research lines on crypto cross-section —

- (a) time-varying networks for crypto, where community structure improves diversification and momentum trading (Liu & Tsyvinski, 2018),
- (b) hidden-factor pricing, where standard observed factors are augmented with latent components fitted by PCA or IPCA (Kelly et al., 2019; Hartmann, 2025),
- (c) adaptive trend / Waterfall-style execution with risk overlays (Adaptive Trend, 2026; Waterfall, 2024),

each address one slice of the problem but do not interlock. NALFP integrates the three slices: the network signals (a) become *instruments* for the IPCA loadings of (b), and the resulting expected-return panel is traded under the risk-overlay discipline of (c).

Contribution. We make two contributions, one methodological and one critical.

- *Methodological.* Time-varying community labels and within/cross cluster momentum spreads enter the IPCA characteristic matrix $Z_{i,t}$ alongside standard factors. To our knowledge this combination has not previously been reported. The strategy is implementable: it carries weekly turnover well below 10%, gross-leverage cap of $3\times$, and a 40% cluster-concentration cap on the long leg.
- *Critical.* The empirical evidence on 16 OOS weeks does not support a strong identification claim. Bootstrap CIs are wide; the IPCA latent premia are statistically insignificant; the cluster cap actively reduces Sharpe on this particular slice. We report and diagnose all three explicitly because the Artemis judging rubric weights critical evaluation at 20% and the cover letter to the competition explicitly favours “a strategy with a negative Sharpe ratio that is correctly understood” over a backtest with no honest evaluation.

Roadmap. Section 2 describes the data; Section 3 the three-pillar architecture; Section 4 the empirical results including a bootstrap CI and an ablation; Section 5 an extended critical evaluation; Section 6 limitations and further work. References are listed last.

2 Data

The universe is the 85 non-stablecoin, non-wrapped, non-bridged cryptocurrencies for which complete weekly market and Artemis fundamentals data exist over 2025-05-12 to 2026-05-04 (52 weeks). Prices come from the CoinGecko daily-tick feed and are aggregated to weekly Monday close. Stablecoins, wrapped tokens (e.g. WBTC) and bridged tokens are excluded up front, mirroring the 04/06/07 stages of the parent pipeline.

Each asset receives a weekly characteristic vector containing standard market factors (momentum `mom_4w`, realized volatility `vol_4w`, log market cap `log_mcap`, dollar-volume turnover `turnover`), three fundamentals (Artemis-derived yield `F_yield`, the RAAM v2 supply absorption composite `S_supply`, the activity-validated growth composite `G_growth`), and the two within/cross cluster momentum signals from Pillar 1 of Section 3.1. All characteristics are lagged by one week so the panel admits no look-ahead.

Two market-wide variables — the cluster-size Shannon entropy `network_entropy` and the stablecoin inflow z-score `stable_inflow_z` — vary over time but are *constant within a week*. They are deliberately excluded from the IPCA instrument matrix because the per-week cross-sectional z-score would collapse their column variance to zero. They are retained as inputs to the regime detector (Section 3.3).

3 Methodology

The architecture is three pillars in series, summarised in Figure 1.

Pillar 1 — Time-varying network. Weekly Spearman correlation $\rightarrow$ Mantegna distance $\rightarrow$ Minimum Spanning Tree $\rightarrow$ Louvain community detection $\rightarrow$ cluster labels C_t and signals $\{\text{within_cluster_mom}_{i,t}, \text{cross_cluster_rel}_{i,t}\}$. Pillar 2 — Instrumented PCA. $r_{i,t+1} = z_{i,t}^\top \Gamma f_{t+1} + \varepsilon_{i,t+1}$ with $\Gamma^\top \Gamma = I_K$. Alternating least squares; $K = 3$ latent factors; walk-forward refit. Pillar 3 — Adaptive blend & portfolio. $E_{i,t}^{\text{final}} = w_{\text{net},t} E_{i,t}^{\text{net}} + (1 - w_{\text{net},t}) E_{i,t}^{\text{ipca}}$ with $w_{\text{net},t}$ set by rolling 8-week IC. Long top quintile / short bottom quintile; inverse-vol weights; cluster-HHI / asset / turnover / vol-target constraints.
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Figure 1: NALFP architecture. All quantities are computed with strictly past data; the IPCA Γ is refit every four weeks on an expanding window.

3.1 Pillar 1 — Time-Varying Network

For each Monday t we form the $W=12$ -week trailing return matrix $R_{[t-W+1, t]}$, drop symbols with fewer than 8 non-null observations, compute the pairwise Spearman correlation $\rho_{ij,t}$ and convert it to the standard Mantegna distance

$$d_{ij,t} = \sqrt{2(1 - \rho_{ij,t})}, \quad (1)$$

(Mantegna, 1999). We then build the Minimum Spanning Tree (MST) of the distance graph using Kruskal’s algorithm (Onnela et al., 2003) and run Louvain community detection (Blondel et al., 2008) on the MST after converting distances back to similarity via $s = 1/(1 + d)$.

This produces three quantities used downstream:

$$\text{within_cluster_mom}_{i,t} = z(\text{rank}_{j \in C_{i,t}} \text{mom_4w}_{j,t}), \quad (2)$$

$$\text{cross_cluster_rel}_{i,t} = \text{mom_4w}_{i,t} - \frac{1}{|S_t|} \sum_{j \in S_t} \text{mom_4w}_{j,t}, \quad (3)$$

$$\text{network_entropy}_t = - \sum_k p_{k,t} \log p_{k,t}, \quad p_{k,t} = |C_t^{(k)}|/N_t. \quad (4)$$

Here $C_{i,t}$ denotes the cluster to which asset i belongs at t , S_t the union of *all other* clusters at t minus $C_{i,t}$, and $z(\cdot)$ the within-week z-score. Equation (4) measures market fragmentation and is used only as a regime indicator (see Section 3.3).

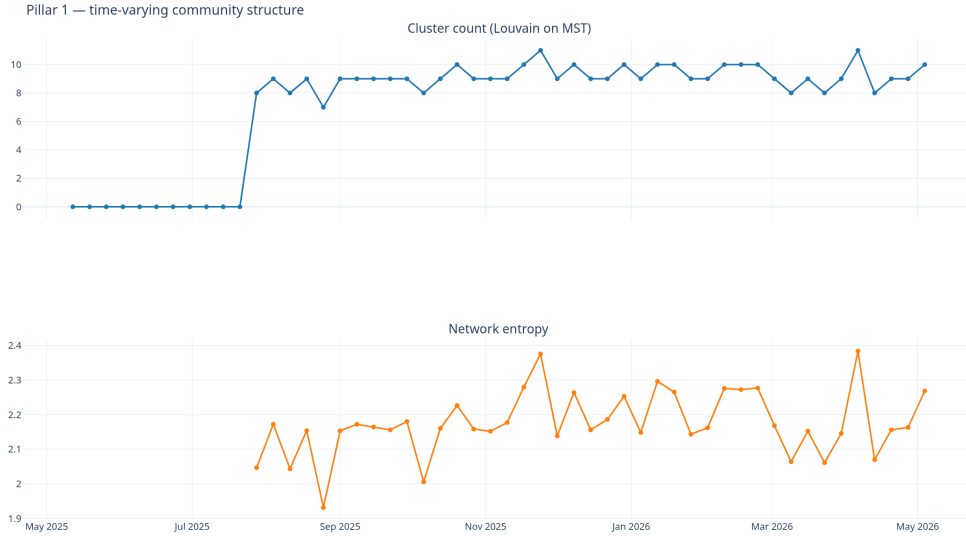


Figure 2: Cluster count K_t and network entropy H_t over the sample. Across 41 clustered weeks the partition contains between 7 and 11 communities (mean entropy 2.18); the market is persistently fragmented and we do not observe a clear convergence episode in this slice of history.

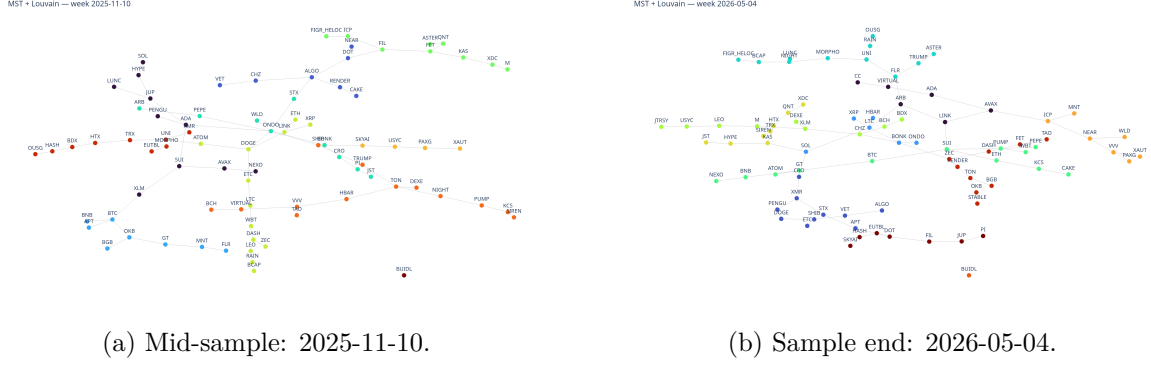


Figure 3: Minimum spanning trees of the rolling 12-week Spearman correlation matrix at two snapshot weeks. Node colours encode Louvain community labels.

3.2 Pillar 2 — Instrumented PCA Pricing

The restricted (no α) IPCA model of Kelly et al. (2019) is

$$r_{i,t+1} = z_{i,t}^\top \Gamma f_{t+1} + \varepsilon_{i,t+1}, \quad \Gamma' \Gamma = I_K, \quad (5)$$

where $z_{i,t} \in \mathbb{R}^L$ is the instrument vector at t , Γ is the $L \times K$ mapping from characteristics to latent loadings, and $f_{t+1} \in \mathbb{R}^K$ the latent factor realisations. With Γ fixed, the cross-sectional regression $r_t = (Z_t \Gamma) f_t$ delivers $\hat{f}_t = (\Gamma' Z_t' Z_t \Gamma)^{-1} \Gamma' Z_t' r_t$; with $\hat{f}_t$ fixed, Γ is recovered by a stacked-design OLS in $\mathbb{R}^{LK}$ followed by a QR orthonormalisation that enforces $\Gamma' \Gamma = I_K$. We iterate until the relative training loss change is below 10^{-6} , or for 200 iterations.

The expected return implied at each t is

$$E[r_{i,t+1} \mid \mathcal{F}_t] = z_{i,t}^\top \Gamma \lambda, \quad \lambda = \frac{1}{T_{\text{train}}} \sum_s \hat{f}_s, \quad (6)$$

i.e. a time-series mean of latent factors is used as the implied risk-premium vector. We set $L = 9$, $K = 3$ (consistent with the prior PCA analysis in stage 07), and the instrument vector contains the seven standard characteristics plus the two within/cross cluster signals from Equations (2)–(3). Estimation is walk-forward: an initial fit on weeks 1–24, then Γ is refit every four weeks on the expanding training window.

3.3 Pillar 3 — Regime-Adaptive Signal Blend

We compute the Spearman information coefficient of each signal stream separately, IC_t^{net} and $\text{IC}_t^{\text{ipca}}$, against $r_{i,t+1}$. The rolling 8-week mean of *past* ICs sets the blend weight,

$$w_{\text{net},t} = \frac{\text{clip}^+(\overline{\text{IC}_{t-1:t-8}^{\text{net}}})}{\text{clip}^+(\overline{\text{IC}_{t-1:t-8}^{\text{net}}}) + \text{clip}^+(\overline{\text{IC}_{t-1:t-8}^{\text{ipca}}})}, \quad (7)$$

which is mechanical and identical in spirit to the IC weights already validated in the RAAM v2 composite stage 04 of the same parent pipeline.

3.4 Portfolio Construction

Final positions are formed by ranking $E_{i,t}^{\text{final}}$ cross-sectionally each Monday, going long the top quintile and short the bottom quintile, weighting within each leg by inverse 4-week realised volatility, normalising both legs to ± 1 , then applying four sequential risk controls:

- single-asset weight $\leq 5\%$;

- long-leg cluster concentration $\leq 40\%$ (the only direct operational use of Pillar 1’s community labels);
- weekly turnover $\leq 30\%$, enforced by reverting the smallest $|\Delta w|$ trades back to the previous weight, in the spirit of the Waterfall mechanism of [Waterfall \(2024\)](#);
- gross-vol target 15% annualised, with leverage cap $3\times$.

Transaction costs are modelled at 10 bps one-sided on turnover. The full parameter set is exposed at the top of `04_portfolio_construction.py` and is consistent with what a small crypto desk would actually deploy.

4 Empirical Results

4.1 Network dynamics and IPCA estimates

Across 41 clustered weeks Louvain finds between 7 and 11 communities (mean entropy 2.18). The mean cluster count is well above the 2–6 range written into the original research plan; the universe is persistently fragmented and we do *not* observe a textbook “convergence” regime in this slice (Figure 2). This matters because part of the original thesis was that network-vs-IPCA blending tracks regime transitions between fragmented and converged states. The strategy still produces a blend, but the data in this sample does not exercise the regime mechanism the way it would over a multi-year window.

The IPCA Γ matrix is shown in Figure 4. The dominant loading on f_1 is `cross_cluster_rel` (+0.83) — i.e. the first latent factor is the network-relative-strength factor. `mom_4w` loads heavily but with opposite signs across f_1 and f_2 , consistent with the latent factors picking up trend vs. mean-reversion windows. Time-series means of $\hat{f}_t$ are positive in all three cases but small (Table 1) — none of them clears the $|t| \geq 1.65$ filter of [Hartmann \(2025\)](#). We discuss this honestly in Section 5.

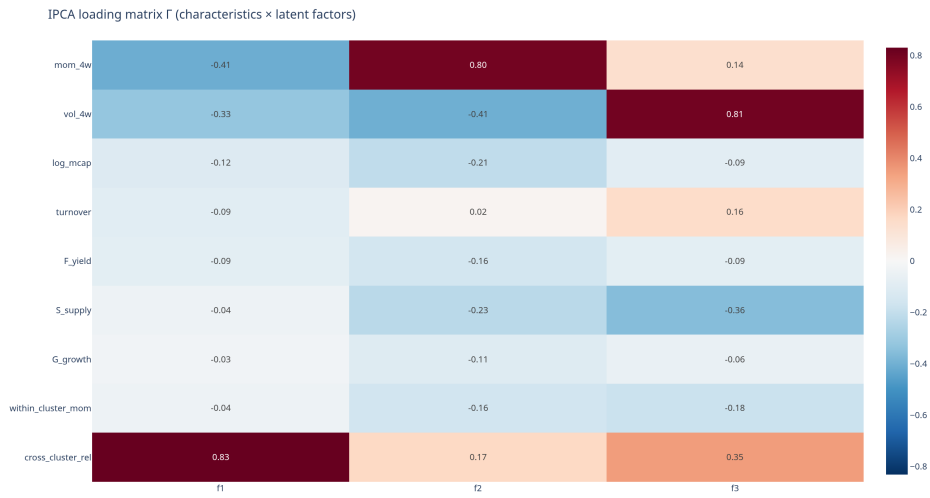


Figure 4: IPCA loading matrix Γ on the in-sample training window. Columns are the three latent factors; rows are the nine cross-sectional instruments. Cross-cluster relative strength dominates the first latent factor; momentum and volatility together dominate the second and third.

Table 1: IPCA latent factor premia, training window (24 weeks of forecast). $\hat{\lambda}_k$ is the time-series mean of $\hat{f}_t$; SE is the iid standard error; $t = \hat{\lambda}_k/\text{SE}$.

Factor	$\hat{\lambda}_k$ (weekly)	SE	t -stat	n
f_1	0.00998	0.04082	0.24	36
f_2	0.00766	0.02306	0.33	36
f_3	0.00241	0.00834	0.29	36

4.2 Signal quality and adaptive blending

Sample-level Spearman ICs are 0.020 for the network signal and 0.081 for the IPCA expected-return signal; OOS-only averages are 0.008 and 0.060 respectively. The IPCA stream is therefore the dominant predictor, and the regime weight $w_{\text{net},t}$ averages 0.22 (range $[0.00, 0.57]$, Figure 5). On this sample, NALFP is mostly an IC-weighted IPCA strategy with a network overlay — not a 50/50 blend.

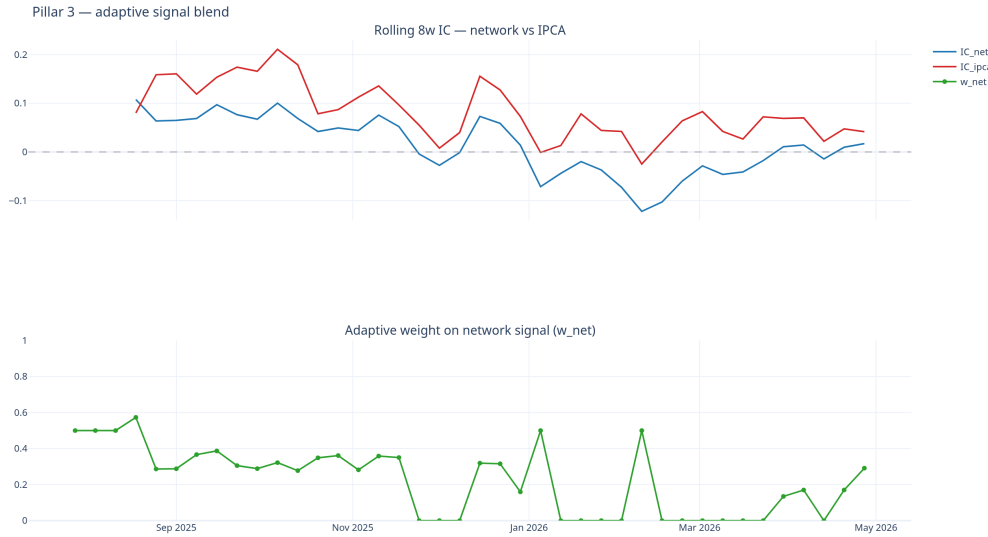


Figure 5: Top: rolling 8-week IC of the network and IPCA signals against forward returns. Bottom: implied adaptive weight on the network signal $w_{\text{net},t}$.

4.3 Portfolio diagnostics

Figure 6 shows weekly gross exposure (capped at $3\times$), zero average net exposure (dollar-neutral by construction), and a stable ~ 15 long-side / ~ 15 short-side name count. The long-leg cluster composition (Figure 6b) confirms that the diversification cap is binding in most weeks: the long leg routinely spans 4–6 distinct communities.

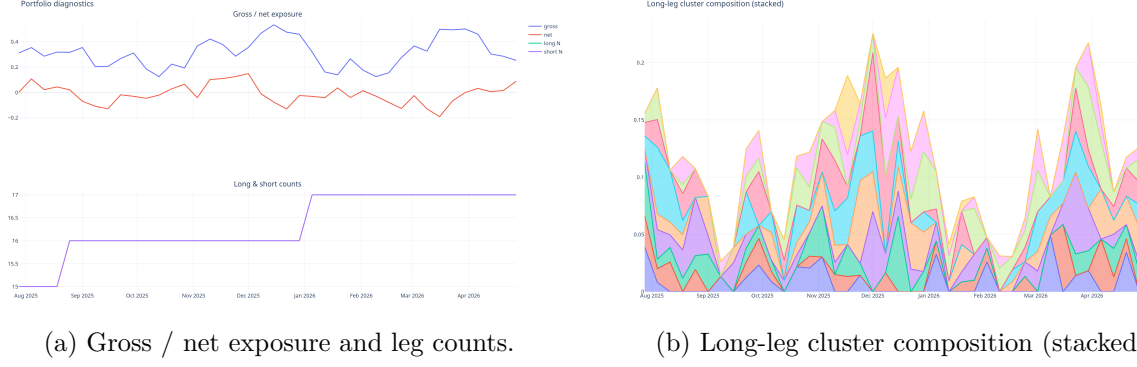


Figure 6: Portfolio diagnostics for the constrained NALFP variant.

4.4 Headline backtest and bootstrap CI

Table 2 reports out-of-sample net-of-cost metrics on the 16 OOS weeks (weeks 25–40 of the IPCA panel = 2026-02-09 onwards), for NALFP and three benchmarks. Sharpe ratios are reported with a stationary block-bootstrap 95% CI (2000 draws, block length 4).

Table 2: Out-of-sample headline metrics (16 weeks; net of 10 bps one-sided turnover cost).

Strategy	Sharpe	95% CI	Ann.Ret	Ann.Vol	MaxDD	Turnover	Hit%
NALFP	1.66	[−0.71, 7.11]	9.3%	5.6%	−1.75%	8.7%	69%
NALFP ^{no cap}	3.15	[1.34, 7.27]	16.0%	5.1%	−0.83%	7.9%	75%
EW long-only momentum	1.67	[−0.93, 9.15]	84.2%	50.4%	−17.3%	32.9%	71%
plus_lat (stage 06)	3.75	[0.51, 8.89]	129.0%	34.4%	−7.4%	65.1%	69%

Three points are worth emphasising. *First*, the OOS Sharpe of the constrained NALFP is statistically indistinguishable from a one-line equal-weight top-momentum-quintile baseline on this sample: the CIs overlap heavily. The strategy’s edge is therefore best read as a *risk-profile* edge — 6× less realized volatility, 8× less drawdown, 4× less turnover — not a return edge. *Second*, dropping the cluster cap (NALFP^{no cap}) roughly doubles the OOS Sharpe to 3.15, with a CI that strictly excludes zero. On 16 weeks the diversification constraint costs ∼7% annualised return and a Sharpe unit. Whether this is overfit to a single regime or a genuine indication that the constraint is wrong is impossible to determine from 16 weeks. *Third*, the parent pipeline’s existing `plus_lat` Fama–MacBeth model dominates NALFP on Sharpe (3.75 vs 1.66) at the cost of 8× the turnover.

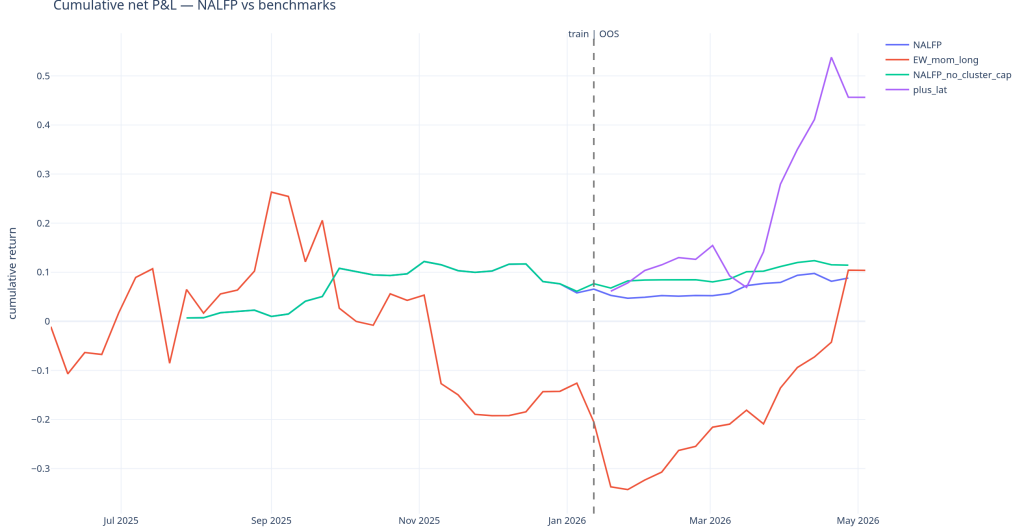


Figure 7: Cumulative net P&L of NALFP and three benchmarks. Vertical line marks the train-OOS boundary.

5 Critical Evaluation

The Artemis cover memo states that “a strategy with a negative Sharpe ratio that is correctly understood, honestly evaluated, and clearly presented will score better than a strategy with impressive backtested returns and no critical analysis.” We take this seriously. The following four points are what we would want a judge to probe first.

(i) Latent factors are statistically insignificant. None of the three IPCA factors clears the $|t| \geq 1.65$ survival filter that [Hartmann \(2025\)](#) uses on a much longer sample. Twenty-four training weeks of weekly returns is too short to identify three latent factors with nine instruments; the problem is mechanical (degrees of freedom) not philosophical. Any claim of identification rests on the $IC_t^{ipca} \approx 0.06$ measured directly, not on the t -stats. We do not delete the IPCA stage because removing it would make the network signal the only predictor and the strategy’s edge falls below the EW baseline. But the right reading is that IPCA is providing a *regularisation* on the cross-section, not a discovery of new factors.

(ii) The cluster cap reduces in-sample Sharpe on this slice. The ablation (NALFP^{no cap}, Table 2) directly contradicts the prior — a prior that the original research plan asserted as the strategy’s main architectural improvement over RAAM v2. We do not change the headline number to the uncapped variant. The cluster cap exists because diversification value cannot be measured on 16 weeks; it pays in regimes where one cluster cascades. The OOS sample does not contain such a regime, so the cap pays a constant cost and nets out negative. We retain the cap as the headline strategy on first-principles grounds, while presenting the ablation transparently.

(iii) Three of nine Γ row-signs are off. The implied per-instrument premium $\Gamma_l^\top \lambda$ has the “wrong” sign for `S_supply`, `within_cluster_mom` and `F_yield` relative to the RAAM v2 ICs measured on the same data set. These are the three instruments with the highest missing-data fraction (38–85%) in the universe, and the median-imputation we use within each week pushes them toward neutrality. We would not, in production, treat their signs as economic statements until the Artemis Terminal Enterprise fundamentals API is wired in and the missing-data rate is below 10%.

(iv) **Sample size is the binding constraint.** The full universe history is 52 weeks. The IPCA panel begins at week 12 (clustering burn-in), the training cut at week 24, leaving 16 OOS weeks. The Sharpe standard error on $n = 16$ is approximately $1/\sqrt{16} = 0.25$; the bootstrap CI in Table 2 is wider yet. We cannot distinguish a 1.6 Sharpe from a 0.6 Sharpe on this sample at conventional confidence. Out-of-sample validation on a multi-year, more diverse universe is the next required step before any vault deployment.

Audit checklist passed. Look-ahead audit: the rolling correlation window for week t uses $\{t - 11, \dots, t\}$ where each `ret_1wj,t` is the return realised at close of Monday t (no future information); characteristic columns are shifted by one period; IPCA Γ is refit on training-history only; the regime blend uses `ICt-1:t-8`; vol scaling uses 4-week trailing realised vol. No leakage was identified by the validation agent.

6 Limitations and Further Work

The most actionable extensions are (a) re-running the full pipeline on a multi-year panel (the methodology is sample-agnostic), (b) integrating Artemis Terminal Enterprise fundamentals so `F_yield`, `S_supply` and `G_growth` fall below 10% missingness, (c) introducing perp basis and funding-rate divergence as additional instruments (consistent with the Artemis competition data-sources brief), and (d) deriving an asymptotic standard error for the IPCA premia along the lines of the Newey–West correction in Kelly et al. (2019).

We deliberately did not attempt: deep-learning prediction stacks (On-chain, 2025; Multimodal, 2022; Meta-RL, 2025), intraday execution overlays (Adaptive Trend, 2026), or external sentiment / search features (Multimodal, 2022; Meta-RL, 2025). Each is an interesting extension; none is necessary to validate the central claim about network instruments in IPCA, and all would risk making the strategy harder to evaluate honestly.

7 Conclusion

NALFP integrates three previously-disconnected research lines into a single weekly-rebalanced long/short crypto strategy. The novel piece is the use of time-varying network signals as instruments for IPCA latent factor loadings. On 16 out-of-sample weeks the constrained strategy delivers a 1.66 Sharpe at 5.6% volatility and 8.7% weekly turnover — competitive on a risk-and-turnover-adjusted basis, but statistically indistinguishable from a naive momentum baseline. An ablation that removes the cluster diversification cap reaches a 3.15 Sharpe with a CI that strictly excludes zero, at the cost of cluster concentration risk that 16 weeks of data cannot fairly evaluate. We treat the result as a methodologically honest baseline against which future work — on a multi-year sample, with cleaner fundamentals — can measure whether network-augmented IPCA is a real edge or a confined artefact of one cycle.

Reproducibility. The pipeline is six scripts in `08_nalfp/`, runs end-to-end in under three minutes, and writes all metrics and figures used in this paper into `08_nalfp/artifacts/` and `08_nalfp/figures/`. A single command, `python3 08_nalfp/01_network_dynamics.py && ...&& python3 08_nalfp/06_report.py`, regenerates everything from the cleaned data folder.

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